Introduction to Manufacturing

(Automation Inside manufacturing)

Basic Manufacturing Concepts

MANUFACTURING ~ MAKING PRODUCTS

Symtetics way of describing Manufacturing
⇓ FLOW of Manuf.

PRODUCT
Requirements

(What is the Goal of the end product)
↓
understanding on what clients need
satisfy a Requirements (HIGH LEVEL)

Design
of the PRODUCT

from detail Design I consider
All COMPONENTS to be assembled

INDUSTRIALIZATION of Product

starting from
Raw Material / Components

Production

different PRODUCT
= different problems

Product

Types of Manufacturing

BIGGEST CLASSIFICATION

Continuous

at the end I obtain a product in which I don't see "Parts"

The elements composing the final product cannot be easily identified.

The process is not reversible.

Examples: Production of Iron, Paper, Chemicals, Pharmaceuticals, Food, …

this Processes have different problems...

Discrete

PART

distinguish different parts of the product

Finite number of components.

Two main phases:

- **Fabrication phase**: set of processes for modifying shape, dimensions, and surface state of a single part.
- **Assembly phase**: set of processes for joining single parts to obtain a (assembled) product.

Examples: Production of Cars, Computers, Appliances, Shoes, Toys, Phones, …

DISCRETE MANUFACTURING

Juicy Salif

Alessi

Design by Philippe Starck

Product and component are the "same"..

JUST ONE complex shape component obtained by different steps of production

DISCRETE MANUFACTURING

Tolomeo

Artemide

Design by Michele De Lucchi

Composed by more components!

(POLYMER based like wires and others...)

↓

Processes to produce each component (Manufacturing)

Many with different processes in different materials

DISCRETE MANUFACTURING

↑

REQUIREMENTS for each component

AW101

Leonardo

Helicopter

COMPLEX

Discrete

Manufacturing

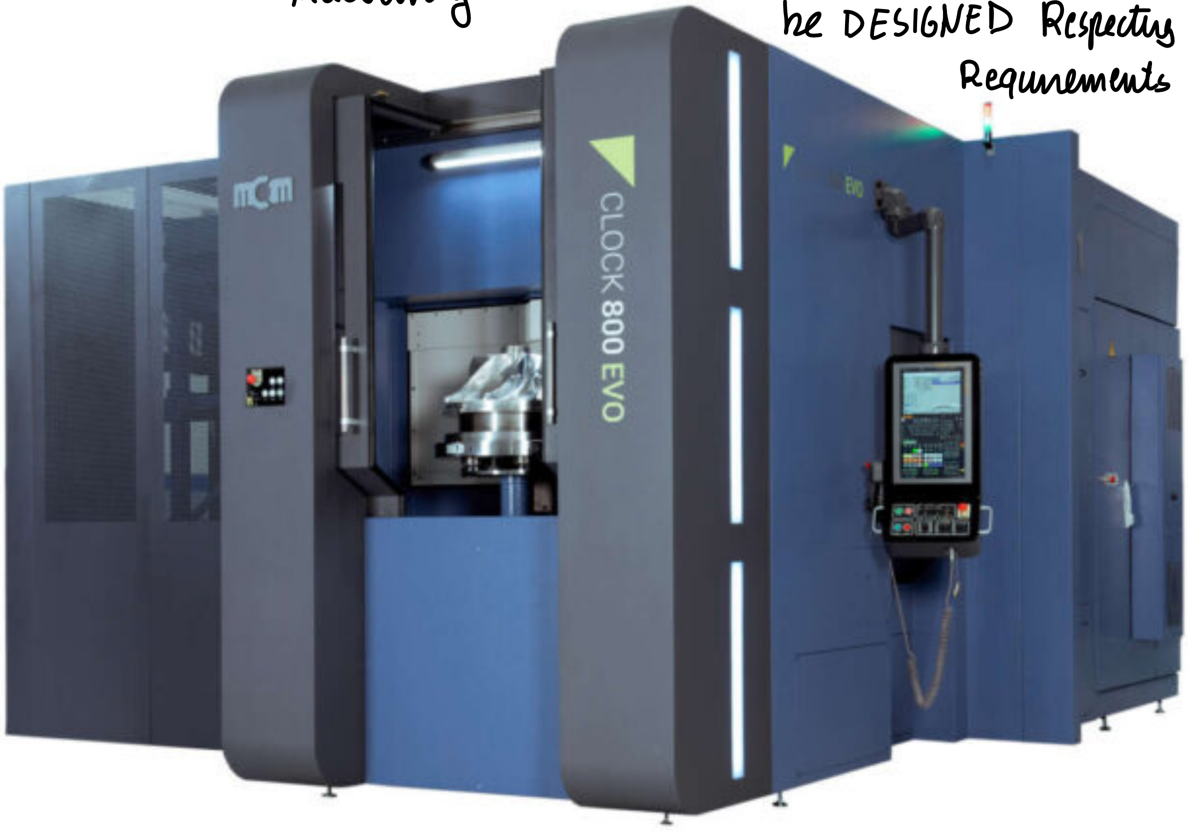
MANY complex

shape components

toghether

Discrete Manufacturing

Machining center → it is a Product to be DESIGNED Respecting Requirements



Clock EVO

Machining Center

MCM spa

Machine tool, PRODUCT composed of PARTS!

DISCRETE MANUFACTURING MACBOOK example

TO produces single components I need PROCESSES → FIRST obtaining the RAW Materials and from the METAL BLOCK and shaping IT, TRANSFORM the shape properly to obtain the elements of the MACBOOK

Machining of Different Types

→ and some FINISHING to meet the Requirements

https://youtu.be/lJx6cF-H__I

← LIST OF KEY WORDS

and CHOICE MATERIAL
Design PRODUCT to be Assembled of PARTS

MULTIPLE PARTS → 1! ⇐ DESIGN FOR MANIFACTURING
WIP PARTS   FINAL PRODUCT   (designed JUST FOR THIS A PRIORI)

DE-MAN
↑ Account the issue of Reusing PARTS (RECYCLE)

⇓
PRECISION
(REQUIREMENTS: TOLERANCES)

MATERIAL → AL

PROCESSES to obtain the FINAL PRODUCT
⇓

- EXTRUSION → FIRST step in which i use raw materials and defining the shape to reach FINAL PART
  ↓ so from the partial product... MODIFING SHAPE (METAL FORMING)
- MACHINING (MILLING)
- LASER MACHINING (unconventional machining)
- GRINDING finishing process of production
- INSPECTION geometrical tollerance → requirements matched?

tollerance = admissible VARIABILITIES { • geometrically • dimensional ?

CONFORMING to Requirement?

{ PROCESSES
MATERIALS
REQUIREMENTS + VERIFICATION

Reffered to all the FLOW of PRODUCTION

A SIMPLIFIED MODEL OF MANUFACTURING PROCESS

DISCUSSING of a single PROCESS → I Refer to a TRANSFORMATION of the PART

"The manufacturing process begins when raw materials are transported to a place – a factory – where labor, know-how, technology, equipment and energy are used to physically and/or chemically **transform** these materials into a product.

CAPABILITY of making products is ESSENTIAL for services!

TRANSFORMATION

single component done in a factory, and PRODUCT Assembled in another

This product is then transported either to another factory, where these same inputs are again applied, or to a place, where those who wish to own it (consumers or other businesses that use the product for their activities) can purchase it."

TRANSFORMATION in different places

FINAL PRODUCT

Globalization

after 2008 manufacturing problems in USA

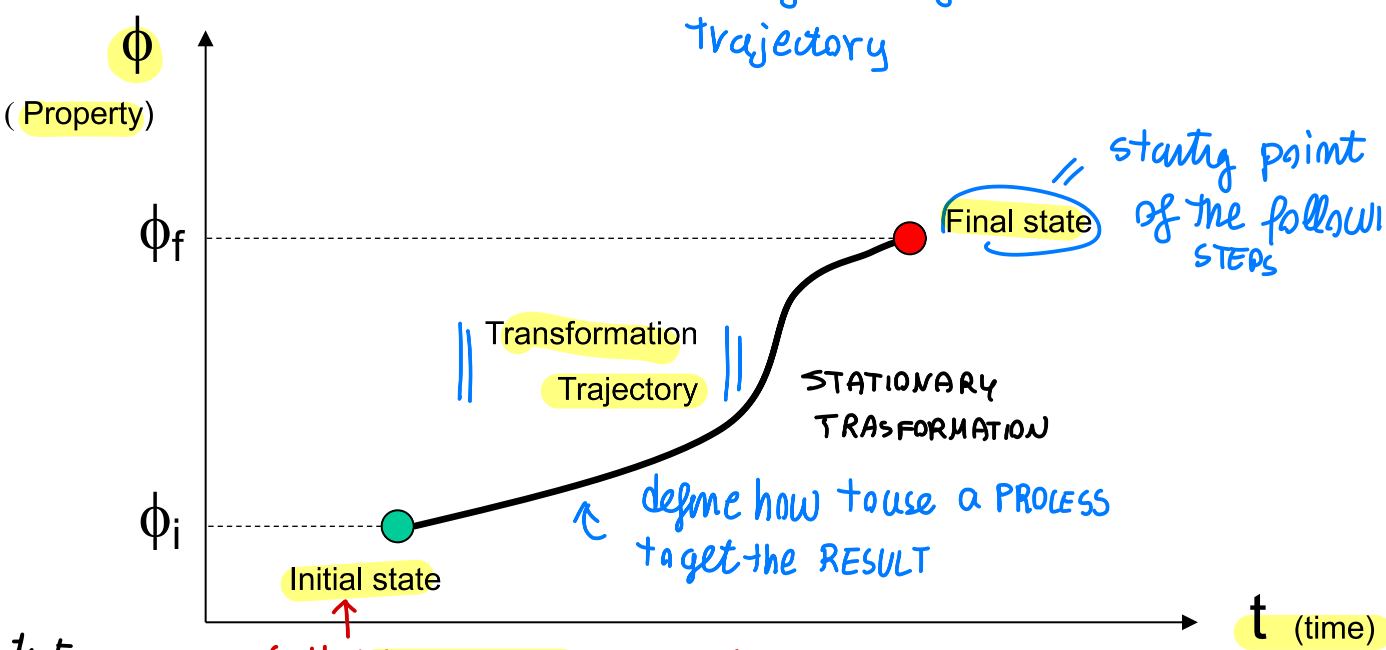
A Framework for Revitalizing American Manufacturing
Barack Obama - December 2009

TRANSFORMATION PROCESS

schematic concept of PROCESS

Modification of a set of product properties obtained by means of selected elementary processes.

MANIFACTURING ~ series of Transformation trajectory



starting point of the following STEPS

STATIONARY TRASFORMATION

define how to use a PROCESS to get the RESULT

with VARIABILITIES in each PART of the SINGLE TRASFORM.

Requirements to be verified cause we are in A STOCASTIC WORD → UNCERTAINTY! (Reduce VARIABILITY?)

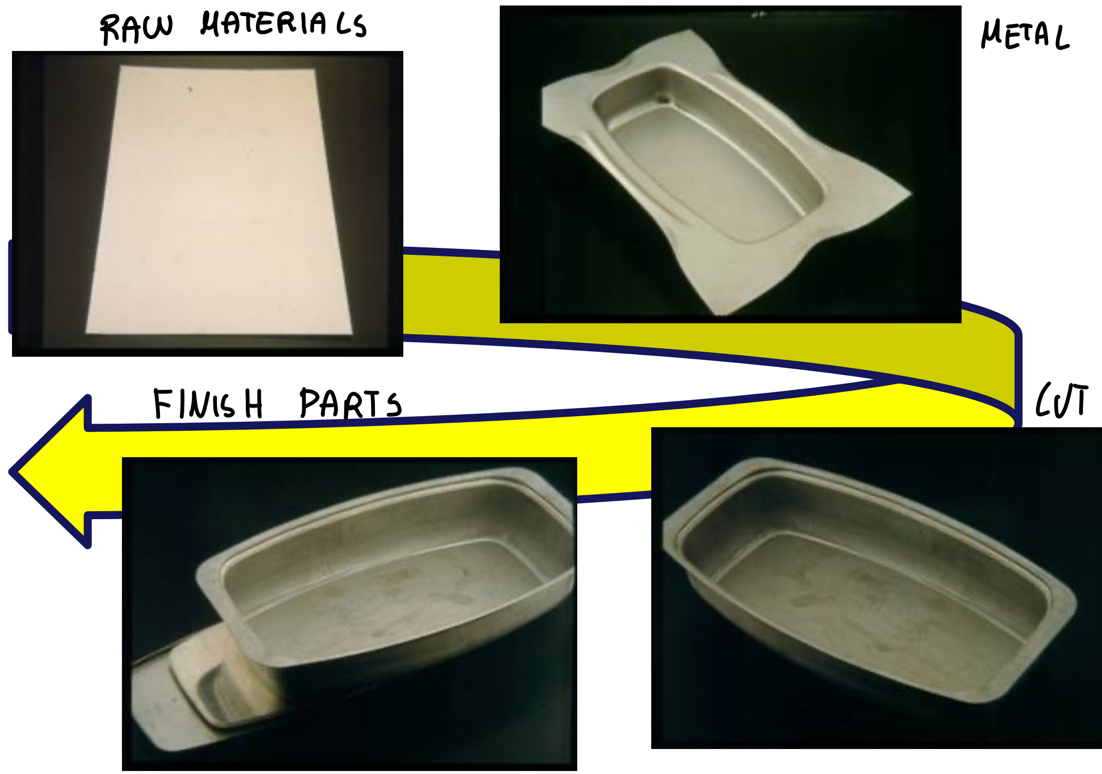
Relevant Transformation EXAMPLE of single product

basic transformation

- Shape and dimension (macro-geometry)
- Surface finish (micro-geometry)
- Mechanical characteristics (stiffness, ductility, etc.)
- State, temperature

(Metal Forming)

Scrap created by Removing METAL PARTS



Fish Kettle
(stainless steel 304, thickness 1,2 mm)

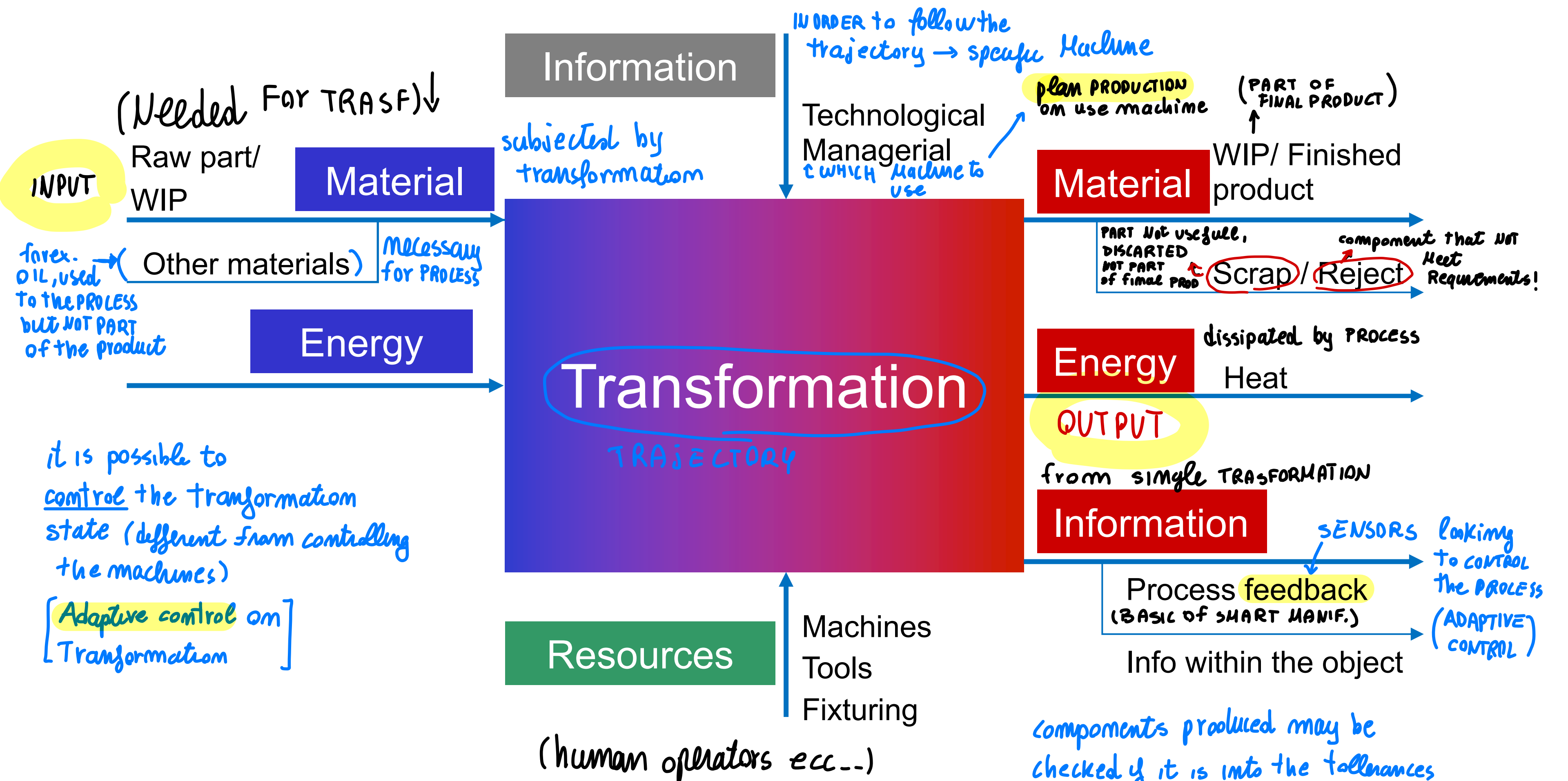
FINAL PRODUCT

(courtesy of Serafino Zani, Lumezzane Gazzolo (BS))

Transformation - Scheme

syntesis of any possible transformation procedure

DIFFERENT KIND of transformation → Represented by a single scheme generally



it is possible to control the transformation state (different from controlling the machines)

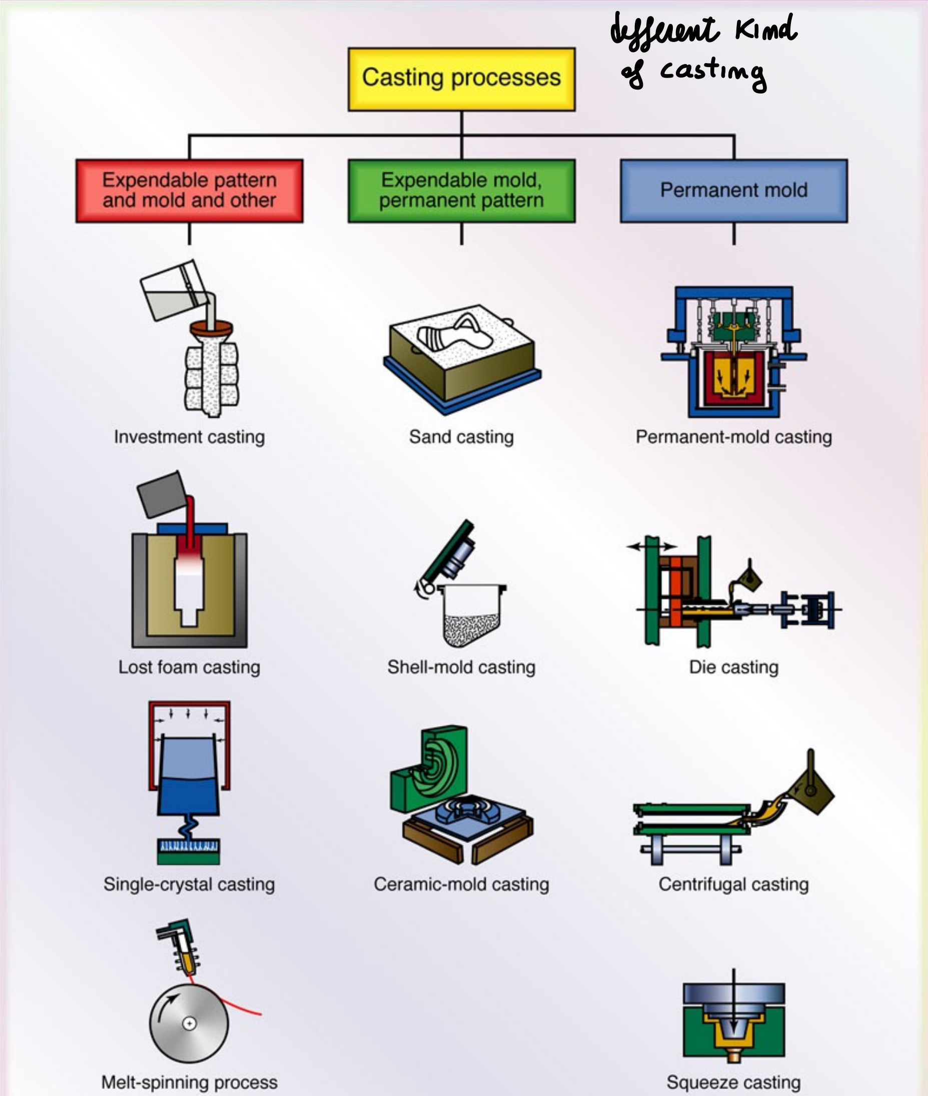
[Adaptive control on Transformation]

components produced may be checked if it is into the tollerances (cause of variability of the process)

METHODS OF TRANSFORMATIONS: CASTING

| Manufacturing process |

different kind of casting



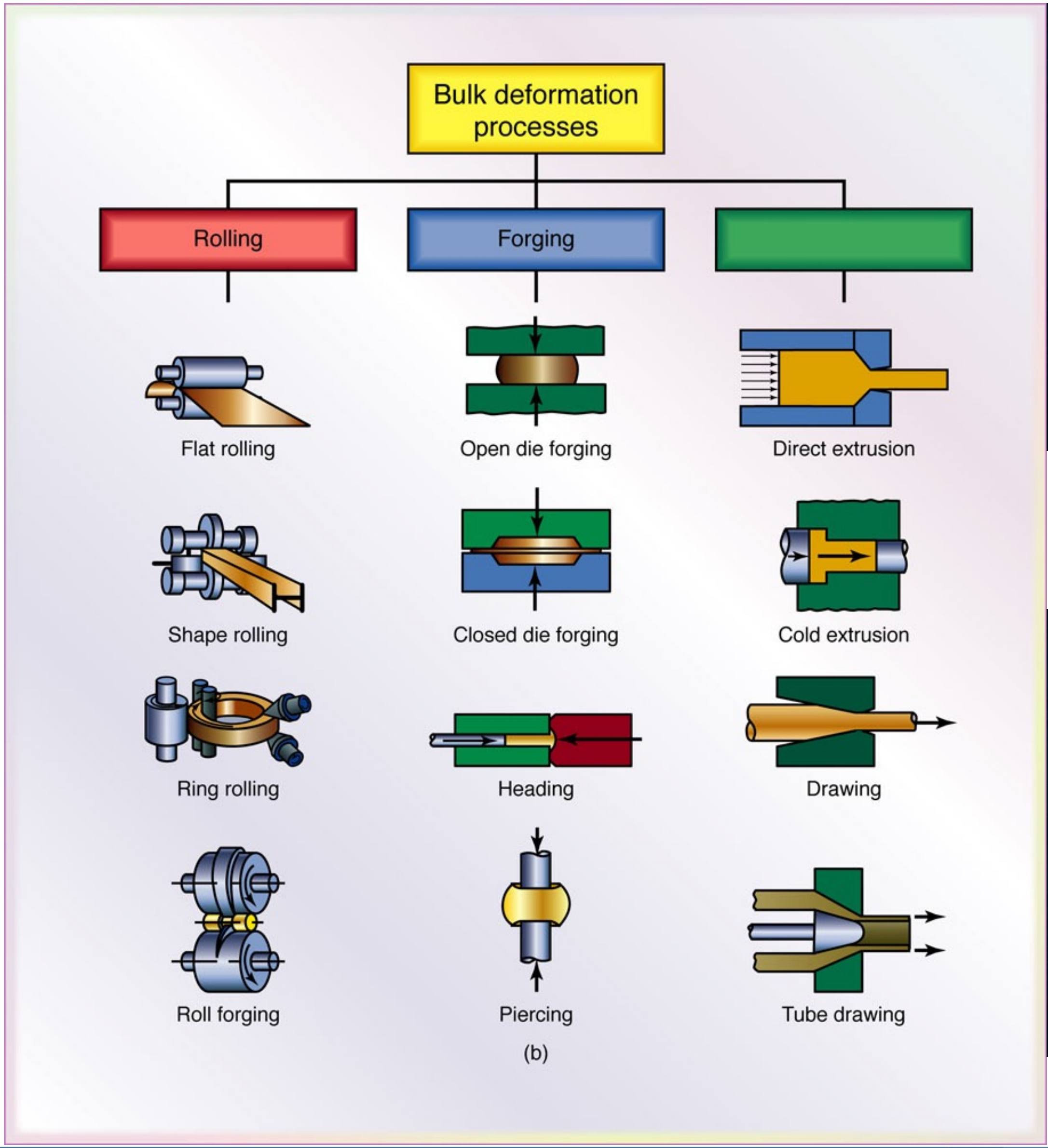
(a)

THE EFFICIENT ENGINEER

CASTING: controlling on the state of the material
(solid, liquid ..)
⇓
prepare the material
Just using temperature to obtain the desired "shell"

CASTING has it´s own problems depending on the method used

Methods of Transformations: Forming and Shaping



THE EFFICIENT ENGINEER

← Primary processes, after the material forming, the first phases

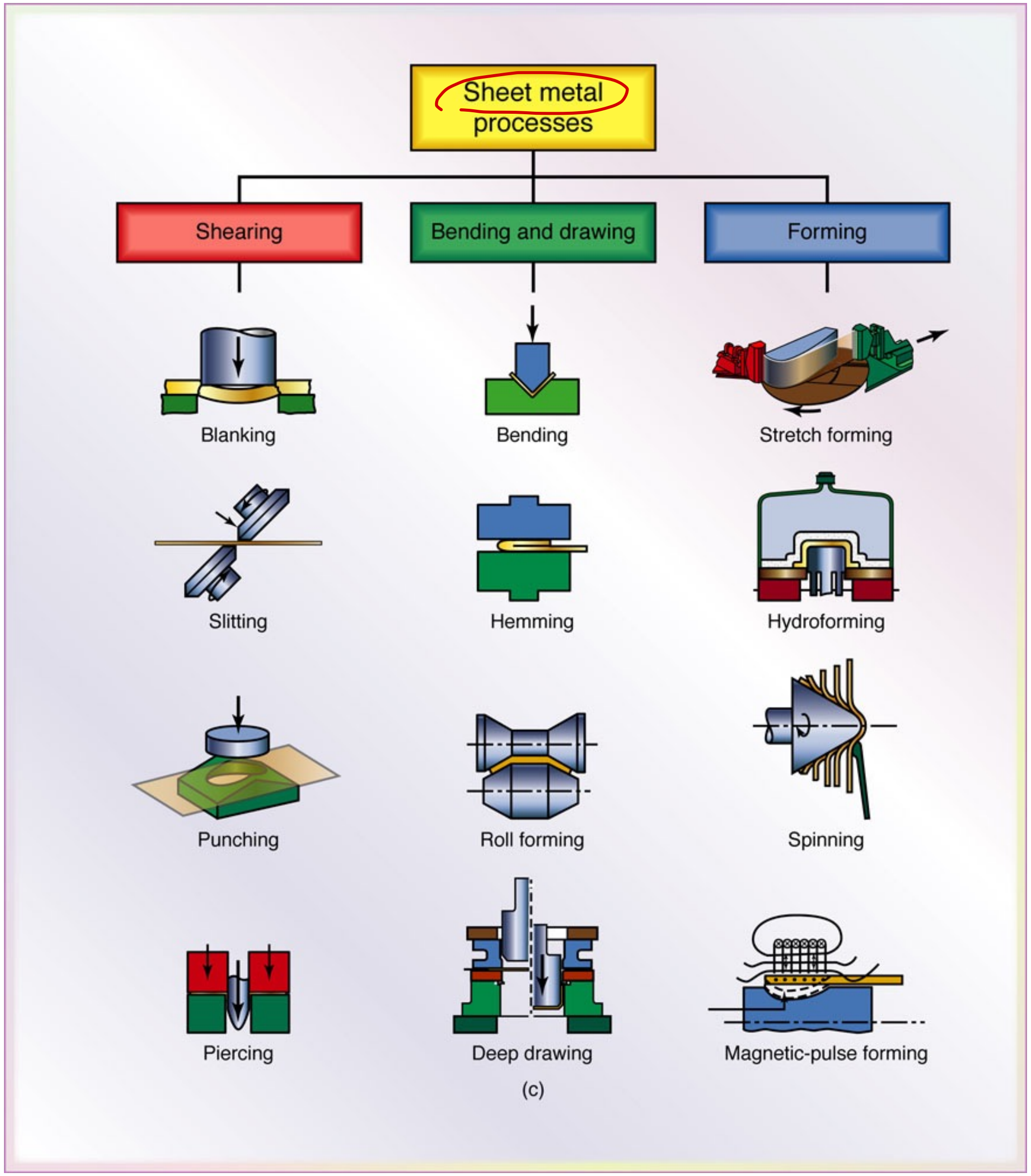
FORGING metal formation using high temperature

ROLLING simple material elaboration

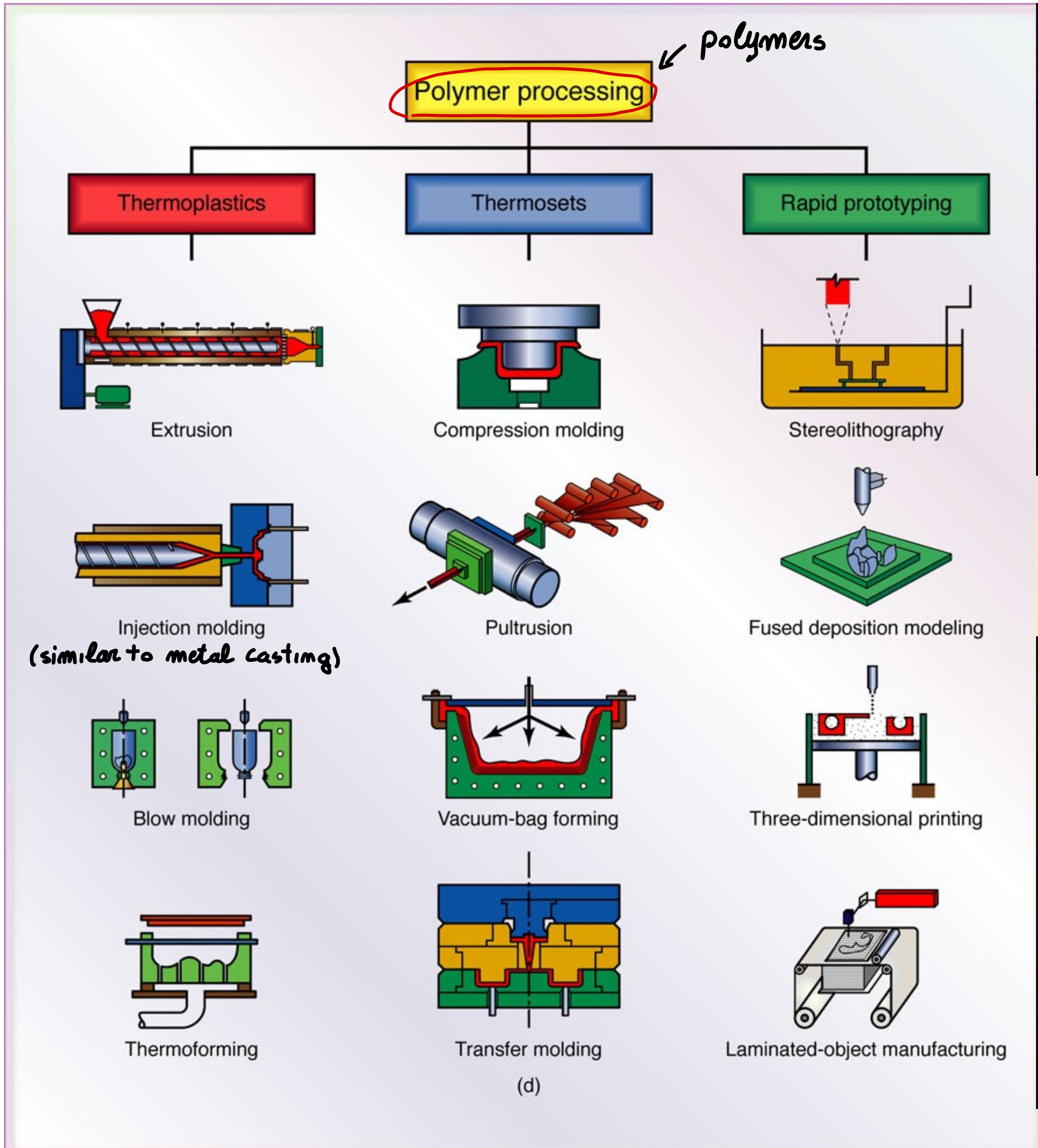
FORMING AND SHAPING → Volume of the material is "CONSERVED"

METHODS OF TRANSFORMATIONS: FORMING AND SHAPING

shaping materials on one dimension



METHODS OF TRANSFORMATIONS: FORMING AND SHAPING

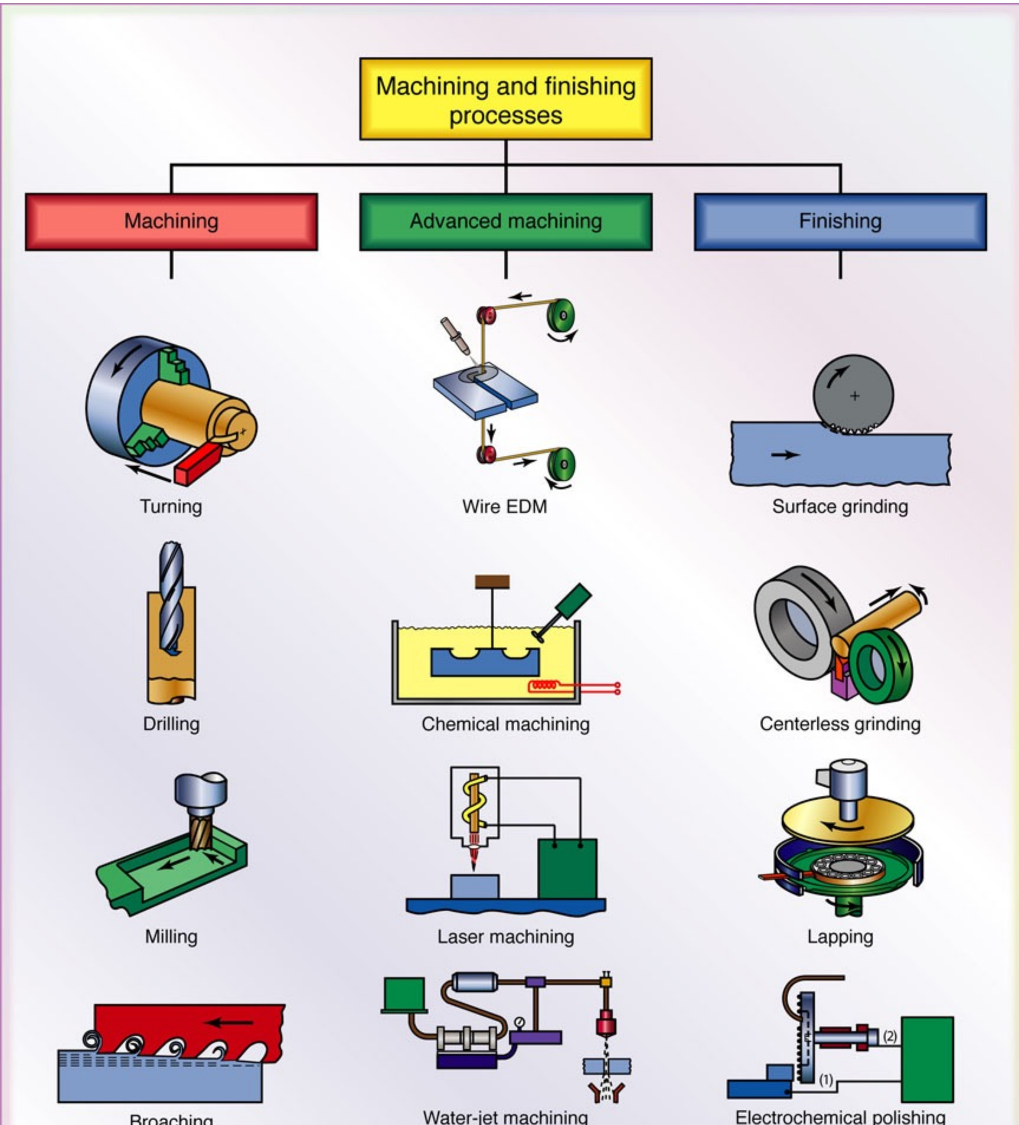


adding materials!
3D PRINTING

THE EFFICIENT ENGINEER

ADDING / REMOVING materials as possible process

METHODS OF TRANSFORMATIONS: MACHINING (Remove material)



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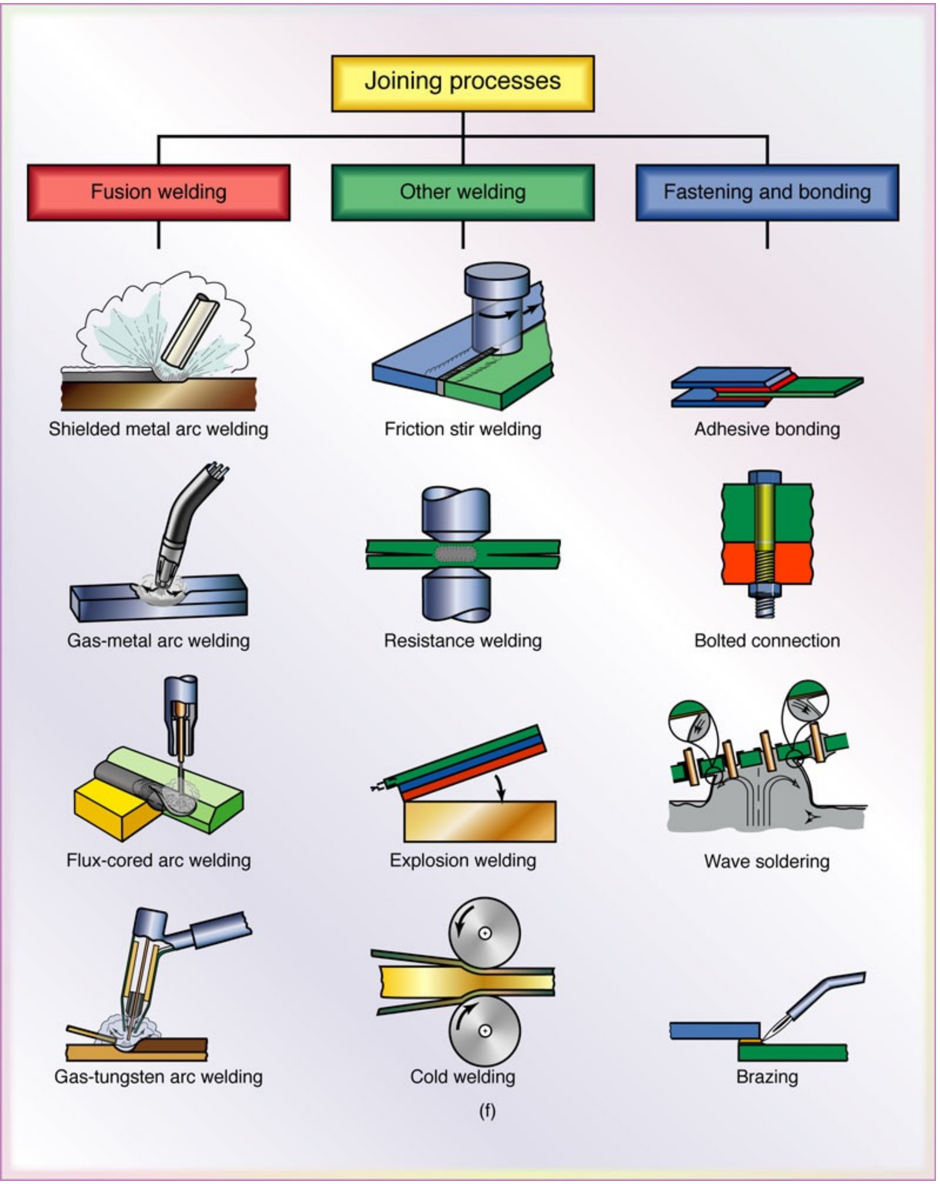
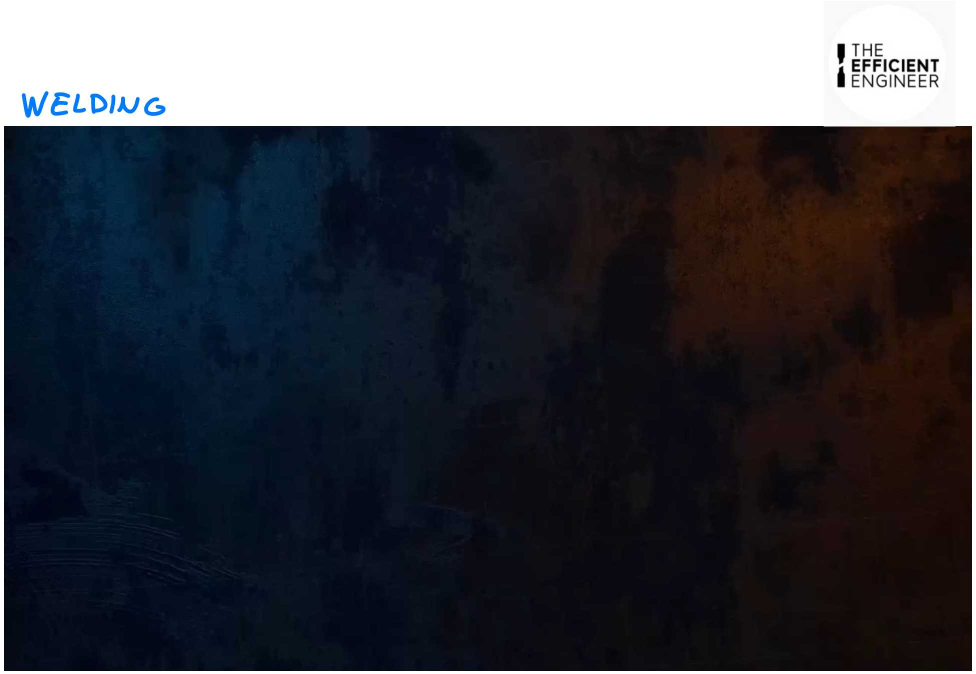
THE EFFICIENT ENGINEER

Methods of Transformations: Joining



WELDING



EXAMPLE – ASSEMBLY

CAR BODY ASSEMBLY ← taking all the component, metal formed

ROBOTIZED $\underbrace{\text{ASSEMBLY CELL}}$

↳ in which each component is
joined together ⇒ position of every single parts

↓

MONITORING THE PROCESS ⇒ measure each component (problem) in order to adjust the overall

smart process: know the shape of each component
to adapt the joining of the parts in a smart way

https://youtu.be/5F6dlts1a34

EXAMPLE – CHAIN OF TRANSFORMATIONS → from row materials to the final part

VERSABILITY of Robot in the production process

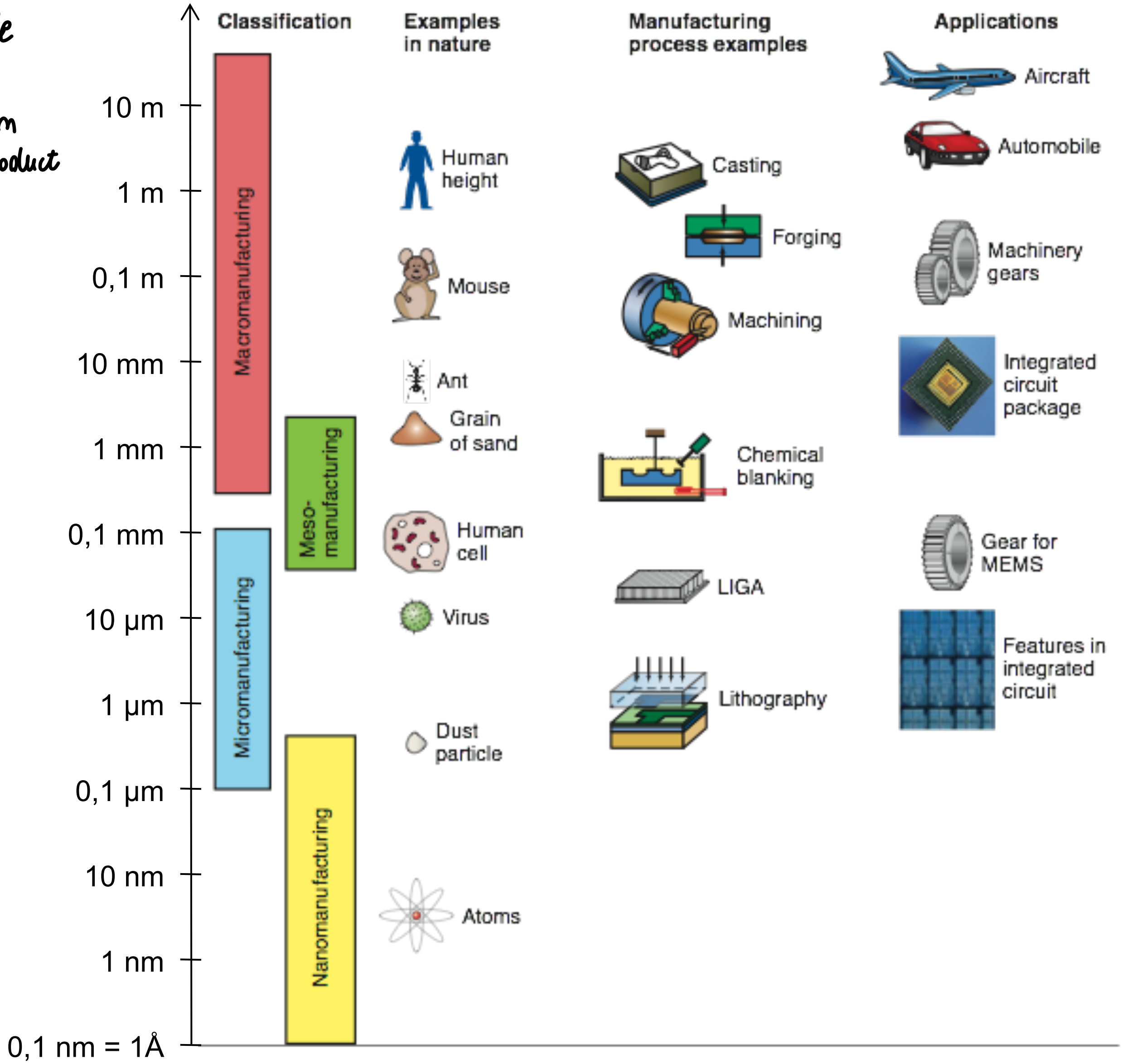
HOW MANY MANIFACTURIN STEPS? } configuring the process's steps

Many different ROBOTS working on the same part together
(with some operation NOT Automated)
↓ machines / Robotics Devices helping the operator in doing the activity

https://youtu.be/8_IfxPI5ObM

SCALES OF TRANSFORMATIONS

there are nanoscale manufacturing process depending on what is the end product



Technical Feasibility

Transformations (manufacturing processes)

should guarantee:

important aspects

↓

- Technical specifications
- Production time
- Environmental regulations
- Operating safety

PARTS OF

Feasibility and

Sustainability

OF THE TRANSFORMATION

TECHNICAL FEASIBILITY

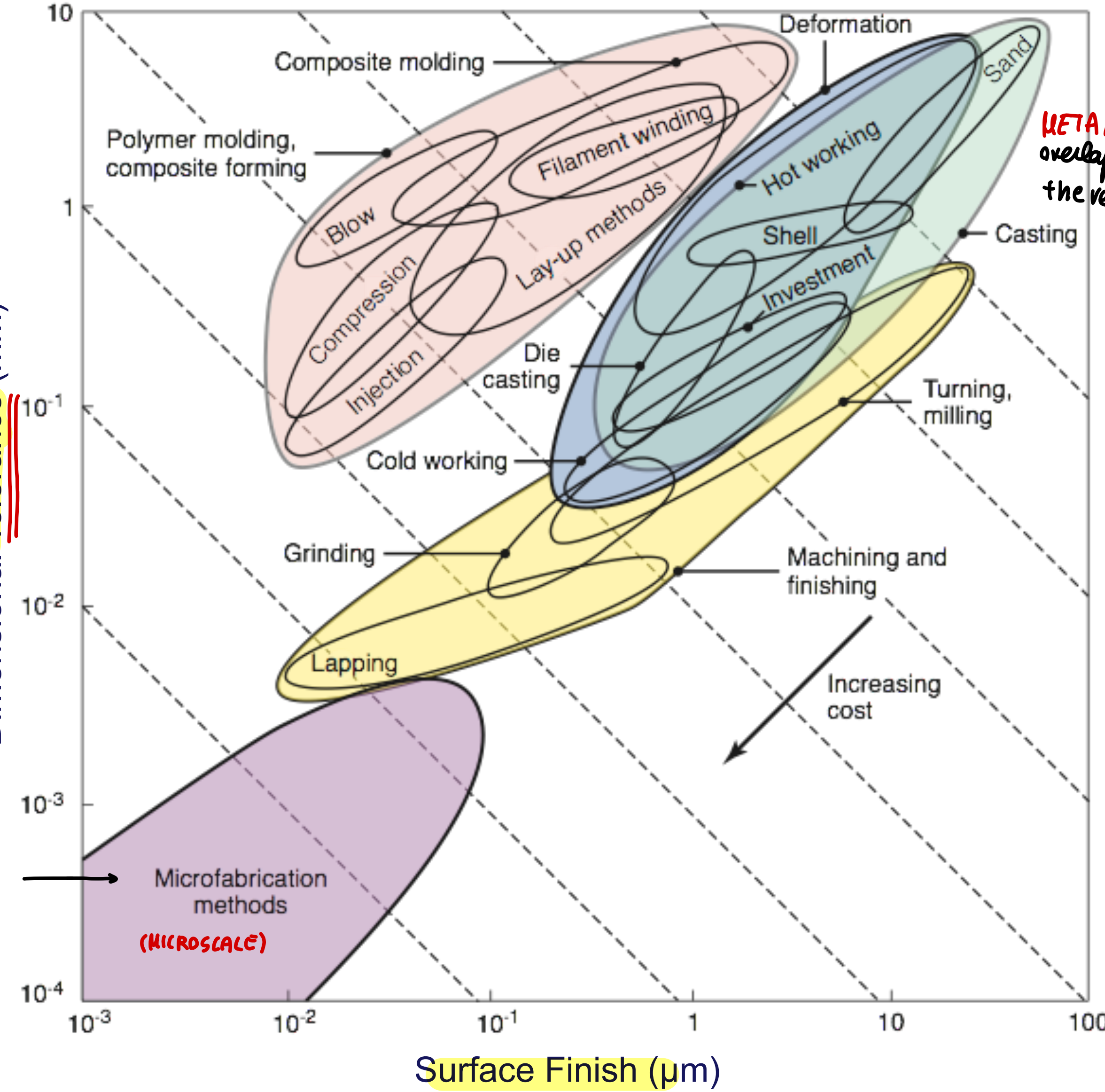
SPECIFICATION of the product

Ashby Chart

we expect dimensional variability on the process ⟹

from 10mm to 10^{-4}mm ←

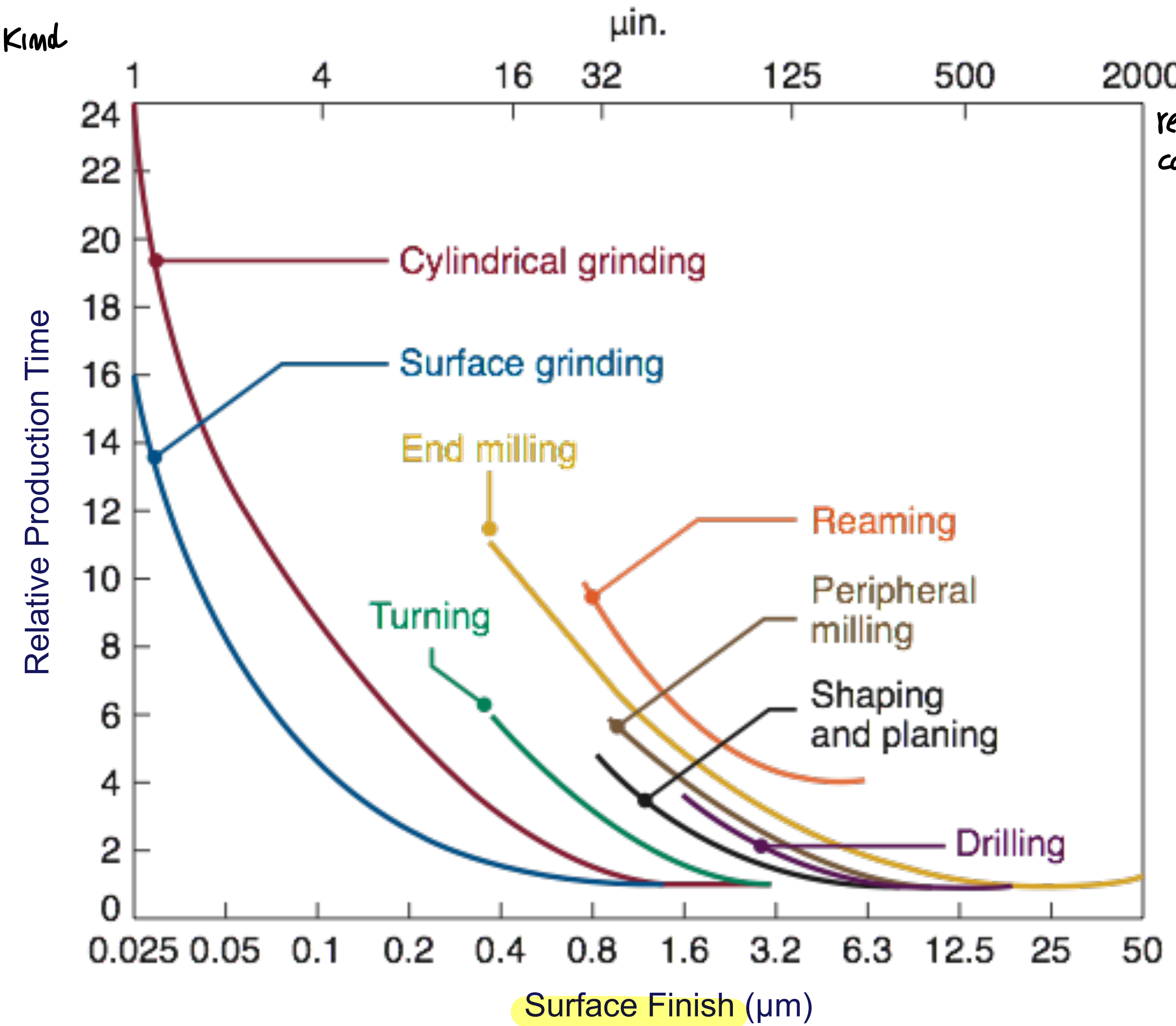
possible area requirements to be respected →



PRODUCTION TIME

In different kind of processes

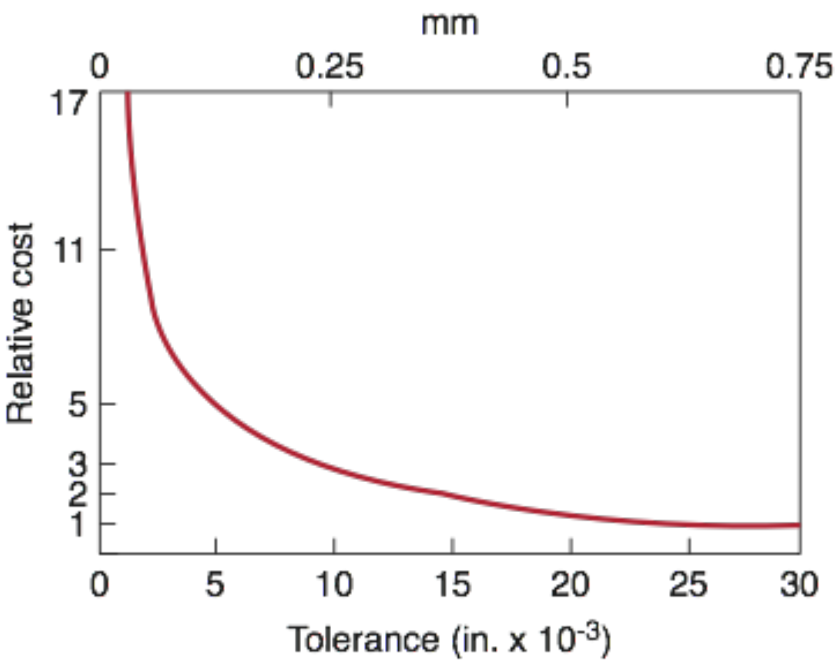
reducing the TIME cause costs



PROFITABILITY

Economical feasibility

cost



Type of machinery	Price range ($000)	Type of machinery	Price range ($000)
Broaching	10-300	Machining center	50-1000
Drilling	10-100	Mechanical press	20-250
Electrical discharge	30-150	Milling	10-250
Electromagnetic	50-150	Ring rolling	>500
Extruder	30-80	Robot	20-200
Fused deposition modeling	40-200	Roll forming	5-100
Gear shaping	100-200	Rubber forming	50-500
Grinding		Stereolithography	80-500
Cylindrical	40-150	Stretch forming	400 - > 1000
Surface	20-100	Transfer line	100 - > 1000
Headers	100-150	Welding	
Injection molding	30-200	Electron beam	75-1000
Jig boring	50-150	Gas tungsten arc	1-5
Horizontal boring mill	100-400	Laser beam	60-1000
Flexible manufacturing system	> 1000	Resistance, spot	20-50
Lathe	10-100	Ultrasonic	50-200
Automatic	30-250		
Vertical turret	100-400		

Note: Prices vary significantly, depending on size, capacity, options, and level of automation and computer controls.

CHOICE OF MANUFACTURING PROCESS

we need to put toghether this two aspect!

Knowing the cost of production cost and how much I earn

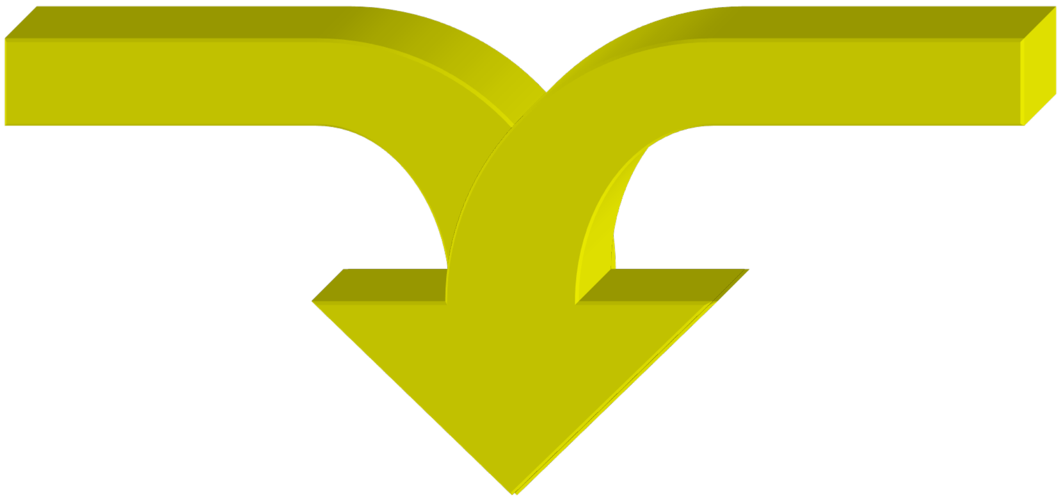
cost - earn = gain

Technological feasibility

valid manufacturing systems for what I have to do

Profitability

(PROFIT RATE = profit for unit of time)



Manufacturing Process

Learning Targets

• Issues of automating the process and all problems of production

Requirements

Design

important for RAW MATERIALS process

(Material Behaviour)

Product requirements (geometrical variation)

Process Planning

CAD/CAM/CAPP = tools helping on the design step

Mechanical Behaviour

Production

(it meets requirements?)

Raw Material / Components

Material Behaviour

Manufacturing Processes

CNC Machine Tools

Smart Manufacturing

Product

Industrial

Geometrical

Metrology